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Microstructural Evolution and Mechanical Properties of Nb-Containing Medium-Mn Low-Density Steels

Chen Ma, Jianlei Zhang, Fenghui An, Gang Chen, Junye Zhou, Changjiang Song, and Qijie Zhai*

Microalloying of Nb has been commonly utilized to enhance the structure and mechanical properties of advanced high-strength steel (AHSS), but its application in medium-Mn low-density steel has been relatively understudied. This study aims to investigate the evolution of nano-scale precipitates and mechanical properties of Fe-12Mn-9Al-3Cr-1.4C-0.02/0.04Nb low-density steels. The findings reveal that the austenite grain size of 0.04Nb steels is approximately half that of 0.02Nb steels due to the precipitation of NbC particles. Moreover, the addition of Nb is found to elevate the formation energy of κ-carbide, thereby impeding its growth and coarsening. Consequently, the intragranular κ-carbides in 0.04Nb steels are consistently smaller than those in 0.02Nb steels, with no coarsened intergranular κ-carbides detected in either steel variant after aging treatment at 723–823 K. Despite a slight reduction on precipitation strengthening of κ-carbides with higher Nb content, the yield strength of 0.04Nb steels exceeds that of 0.02Nb steels by ≈71 MPa, mainly due to additional fine grain strengthening and dislocation strengthening. The strengthening mechanisms in the Nb-containing low-density steels are analyzed quantitatively. The role of Nb in regulating the microstructure and mechanical properties of medium-Mn low-density steels is comprehensively discussed, offering valuable insights for the alloy design of low-density steel.

1. Introduction

Due to the good comprehensive mechanical properties, medium-Mn steels have become highly interesting for automotive structure applications and are recognized as the third-generation advanced high-strength steels (AHSS).[1–4] Through utilizing various strengthening mechanisms and triggering transformation-induced plasticity (TRIP) effect or twinning-induced plasticity (TWIP) effect in retained austenite, the medium-Mn steels could achieve an excellent product of strength and elongation (PSE) up to 80 GPa%.[1,4] Recently, as automobile lightweighting is increasingly valued, a significant amount of Al is added into the medium-Mn steel to develop low-density steel.[5–7] It has been reported that adding 1 wt% Al could decrease the steel density by 1.3–1.5%.[8,9] Moreover, the addition of Al could facilitate the formation of κ-carbide, a nano-scale strengthening precipitate in austenite, thereby substantially improving the mechanical properties of steels.[10,11] Our previous work has reported that, a κ-carbide-strengthened Fe-12Mn-9Al-1.4C-3Cr (wt%) steel, which underwent cold rolling, exhibited an impressive tensile strength of 1.5 GPa while maintaining a total elongation of 15%.[12]

The precipitation behaviors of κ-carbide have a significant impact on the mechanical properties and strain-hardening behaviors of the medium-Mn low-density steels.[6,13,14] The increase in both quantity and size of κ-carbides can enhance their strengthening effect, resulting in a strength contribution of more than 400 MPa.[12] However, Liu et al. reported that aging at 550 °C for 60 min resulted in the formation of a large amount of intragranular and intergranular κ-carbides in Fe-11Mn-10Al-1.2C (wt%) steel, achieving a high yield strength of 1407 MPa but with a low elongation of 7.4%.[6] Additionally, the growth of κ-carbides could hinder dislocation glide and slip band formation, leading to decreased strain-hardening ability.[11,14,15] Compared to high-Mn low-density steels, the regulation of κ-carbides in medium-Mn low-density steels appears to be more challenging. The higher Mn content in steels could slow down the formation rate of κ-carbides,[16,17] which allows for more regulation space, whereas in the medium-Mn low-density steels, it is easier to form coarse κ-carbides.[13] Further exploration of

C. Ma, J. Zhang, J. Zhou, C. Song, Q. Zhai
Center for Advanced Solidification Technology (CAST)
School of Materials Science and Engineering
Shanghai University
Shanghai 200444, P. R. China
E-mail: jianleizhang@shu.edu.cn

J. Zhang
Shanghai Engineering Research Center for Integrated Circuits and Advanced Display Materials
Shanghai University
Shanghai 200444, China

F. An
Jiujiang Engineering Research Center of Advanced Welding
Jiujiang University
Jiujiang 332005, P. R. China

G. Chen
Ningbo Branch of Chinese Academy of Ordnance Science
Ningbo 315103, P. R. China

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regulatory strategies for nano-precipitation behaviors and mechanical properties in medium-Mn low-density steels is necessary.

The addition of microalloying elements, such as Ti, V, and Nb, has been shown to effectively control the microstructure and performance of medium-Mn steel.[18–20] A study by Park et al. demonstrated that the addition of 0.2 wt% V could trigger the formation of VC precipitates in a Fe-8Mn-0.2C-3Al (wt%) steel, leading to significant grain refinement in austenite and a strength enhancement of 246 MPa.[18] Similar effects on grain refinement and strength enhancement also observed in Ti-containing and Nb-containing medium-Mn steels, due to the formation of TiC and NbC particles.[19,20] The addition of 0.05 wt% Nb to a Fe-5.6Mn-1.2Al-0.17C steel could refine the austenite grain by forming nano-scale Nb(C, N) particles.[20] Additionally, the smaller average austenite grain caused by the precipitation of Nb(C, N) particles was proven to improve the stability of austenite.[20] However, there are few research on how microalloying elements regulate the microstructure and performance of medium-Mn low-density steels, especially for the precipitation behaviors of κ-carbides.

Park et al. have reported that the addition of 0.1 wt% Nb resulted in the formation of NbC carbides in a Fe-20Mn-8Al-1.0C (wt%) steel, which reduced the C content in austenite, decreasing the precipitation of κ-carbides.[21] The Nb(C, N) carbides typically possess a hard characteristic, contributing to significant strain-hardening potential.[22] But it was questionable whether the limited formation of NbC could significantly reduce the C content in austenite and thus affect the formation of κ-carbides. Consequently, the influence of Nb microalloying on the microstructure and mechanical properties of medium-Mn low-density steel is worth exploring in detail. This work designed two Fe-12Mn-9Al-3Cr-1.4C-0.02/0.04Nb (wt%) low-density steels to explore the microstructural evolution and mechanical properties of the Nb-containing medium-Mn low-density steels. Considering the influence of aging treatment on the nano-precipitation behavior in low-density steels, especially κ-carbides,[8,23] the studied steels were further aged at 723, 773, and 823 K. The microstructure evolution was examined using a scanning electron microscope (SEM) equipped with electron back scatter diffraction (EBSD) system and a transmission electron microscope (TEM). Additionally, the influence of Nb addition on the nano-precipitation behavior was elucidated through thermodynamic calculations and first-principles calculations.

2. Experimental Section

The Fe-12Mn-9Al-3Cr-1.4C (wt%) low-density steels with 0.02 wt% and 0.04 wt% Nb addition in this work were prepared using a centrifugal casting method under near-rapid solidification. Initially, the ingot was prepared by induction melting in a water-cooled copper crucible. Subsequently, the ingot was re-melted and swiftly poured into a rotating copper mold at a high speed to obtain steel strips with dimensions of $60 \times 70 \times 2.5$ mm through near-rapid solidification. Aging treatment was performed on a portion of these steel strips in a tubular vacuum furnace at temperatures of 723, 773, and 823 K for 3 h. Subsequently, an appropriate aging temperature was selected to investigate the structure evolution and mechanical properties of steels under different aging periods. For simplicity, hereafter, the steels containing 0.02 wt% Nb and 0.04 wt% Nb are referred to as 0.02Nb steels and 0.04Nb steels, respectively.

The actual measured chemical compositions of steel strips are presented in **Table 1**. X-ray diffraction (XRD) analysis for the steel strips was conducted using a D/max-2500 instrument at 40 kV, scanning in the range of 30°–100° with a speed of 4°/min. An Apollo 300 thermal field emission SEM equipped with an EBSD detector was used for the microstructure characterization. The scanning step size for EBSD characterization was set as 300 nm. Prior to SEM observation, the samples underwent electropolishing with 10 vol% perchloric acid alcohol solution under 25 V for a duration of 10 s. To analyze the elemental distribution, an electron probe microanalysis (EPMA) was carried out using a JXA-8050G instrument at an electron beam voltage of 15 kV. The TEM characterization for the nano-scale precipitates was conducted using a JEOL JEM-2100 TEM operating at 200 kV. For the preparation of TEM specimen, a double-jet electropolishing technique was utilized with a solution composed of 10% perchloric acid and 90% alcohol acid at a temperature of ≈−30 °C and a voltage of 35 V. The microhardness of the strips was measured using a Vickers hardness tester (MH-5L) with an indentation load of 200 gf and a holding time of 20 s. The tensile property tests were carried out using a MTS Criterion 44 universal testing machine at ambient temperature. The strain rate during the tests was set as $2.5 \times 10^{-4}\ s^{-1}$. The test specimens had a gauge width of 4 mm and a gauge length of 12 mm to ensure consistent measurements. To ensure accurate data analysis, each steel strip underwent a minimum of four repetitions during the testing process.

To explore the influence of Nb addition on the formation of κ-carbide, this study employed spin-polarized density functional theory (DFT) calculations within the generalized gradient approximation (GGA) utilizing the first-principles Perdew–Burke–Ernzerhof (PBE) recipe, which was implemented in Vienna Ab Initio Simulation Package (VASP).[24–27] The ion nuclei were characterized by projected enhanced wave (PAW) potentials,[28,29] while the valence electrons were treated with a plane-wave basis set and a kinetic energy cutoff of 500 eV. To accurately compensate for the partial occupation of the Kohn–Sham orbitals, the Gaussian smearing approach was utilized with a width of 0.05 eV. The convergence of the geometry optimization process was determined by ensuring that the energy change was below 0.03 eV $Å^{-1}$. The U scheme with a value of 2.59 eV was applied for Fe atoms. The slab structure had a vacuum spacing of 15 Å in the direction perpendicular to the structure plane. Brillouin zone integration of the structure was achieved using $6 \times 6 \times 6$ Monkhorst–Pack k-point sampling.

Table 1. Chemical compositions (wt%) of the studied low-density steel strips.

Samples	Mn	Al	Cr	C	Nb	Fe
0.02Nb steel	12.80	9.01	3.01	1.39	0.021	Bal.
0.04Nb steel	12.82	9.12	2.99	1.40	0.039	Bal.

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3. Results

3.1. Structure and Mechanical Properties of Steels

3.1.1. Microstructure Evolution

Figure 1 displays the XRD patterns of the 0.02Nb and 0.04Nb steels, in which the microstructure of all the Nb-containing steels consists of ferrite and austenite. After aging treatments at different temperatures, the diffraction peak intensity of both austenite and ferrite in the 0.02Nb and 0.04Nb steels shows almost no significant change, as depicted in Figure 1a,b.

Figure 2 shows SEM images of the 0.02Nb and 0.04Nb steels aged at different temperatures. It can be found that irregular-shaped ferrite is randomly dispersed within the austenite matrix. In order to investigate the effects of Nb content and aging treatment temperature on the microstructure of low-density steels, an ImageJ software was employed to measure the ferrite content in different samples. During the measurement process, an image selection tool in the software was used to select all irregularly shaped ferrite in SEM images, and calculated the proportion of ferrite phase area in the SEM images as the ferrite content. To ensure the accuracy of statistical data, more than five SEM images were selected for each sample for analysis, and the average value of the measurement results was taken as the ferrite content. The ferrite content in the as-cast 0.02Nb steel is measured as ≈16.2 vol%. Following aging treatment at 723, 773, and 823 K for 3 h, the ferrite content in the steels is 16.4, 17.1, and 18.3 vol%, respectively. Similarly, the 0.04Nb steels under the same aging states have ferrite contents of 16.3, 17.2, 18.4, and 18.1 vol%, respectively. These results indicate that the ferrite contents in the low-density steels are not affected by the aging temperature, which aligns with the observation from XRD analysis. Furthermore, the Nb addition leads to the formation of submicron white particles within the austenite matrix, as

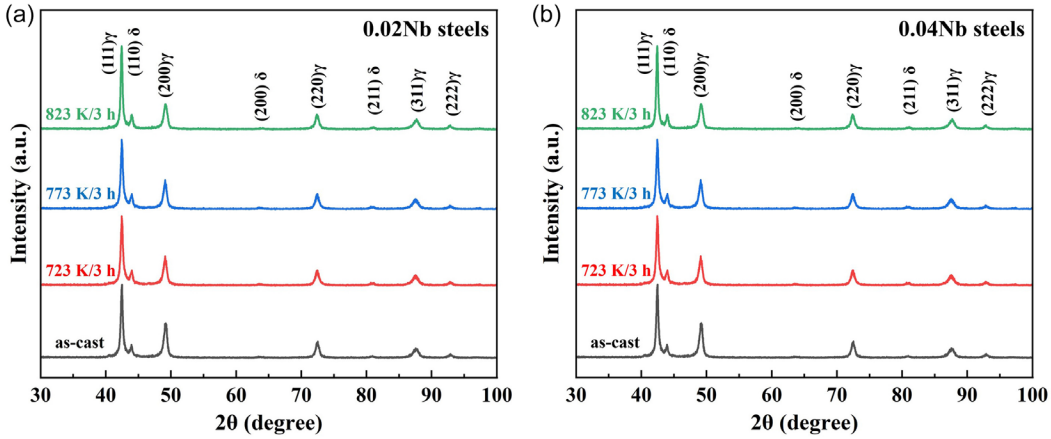


Figure 1. XRD patterns of the a) 0.02Nb steels and b) 0.04Nb steels, at different aging states.

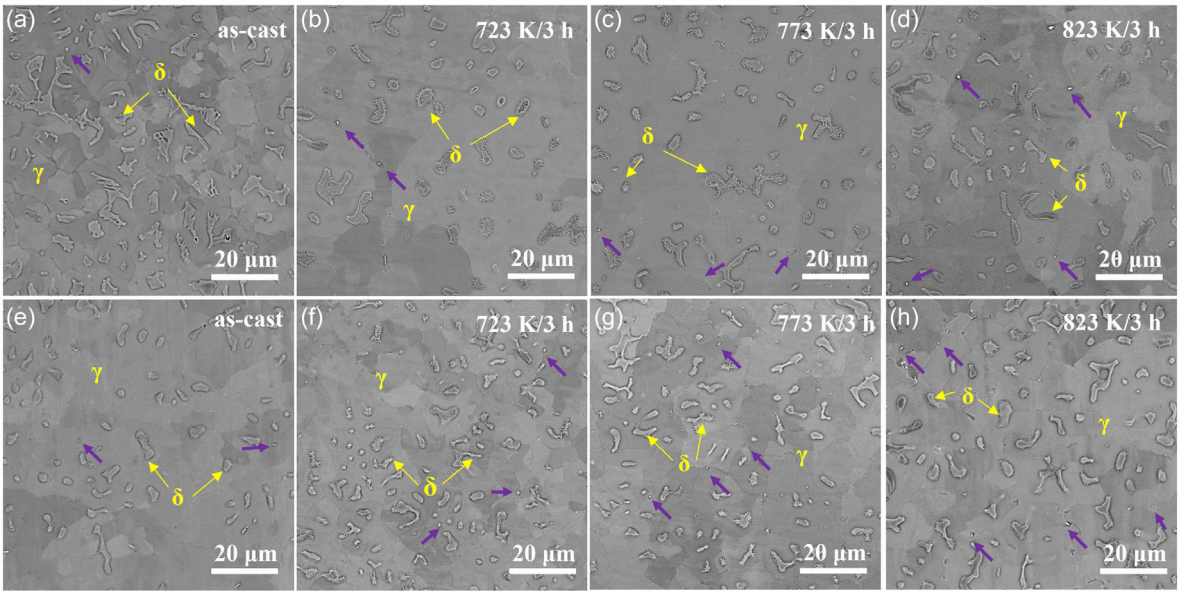


Figure 2. SEM micrographs of the a–d) 0.02Nb steels and e–h) 0.04Nb steels, at different aging states.

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indicated by the purple arrows in Figure 2. **Figure 3** shows the TEM selected area electron diffraction (SAED) analysis and corresponding EDS analysis of the white particles in as-cast 0.02Nb and 0.04Nb steels, in which the white particles are identified as NbC. In order to analyze the evolution of NbC particles in 0.02Nb and 0.04Nb steels, at least five SEM images of each sample were selected for analysis by ImageJ software. According to the statistical results shown in **Figure 4**, both the size and volume fraction of the NbC particles increase slightly with higher aging temperature and more Nb content.

Figure 5 shows EBSD orientation color maps for the Nb-containing steels. Although both ferrite and austenite were present in the EBSD maps, these two phases could be clearly distinguished based on the morphology of ferrite in Figure 2.

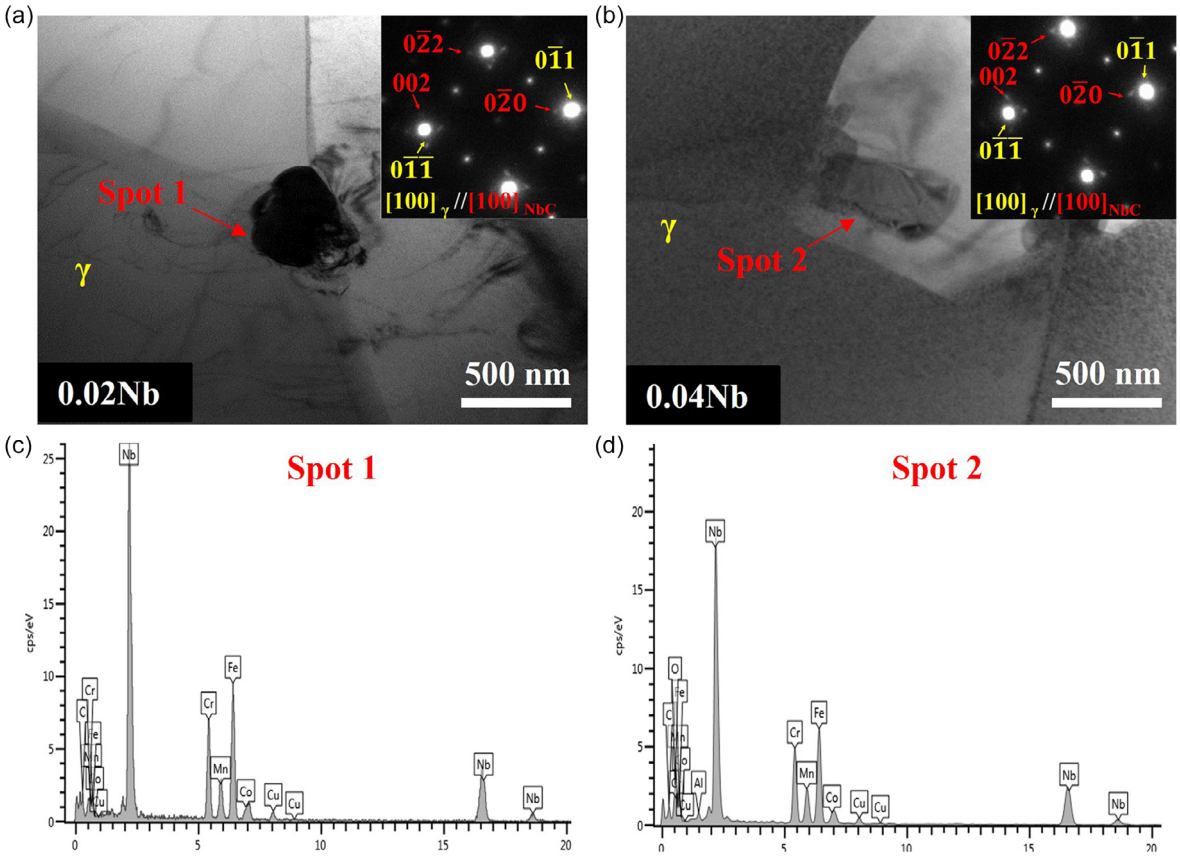


Figure 3. TEM analysis of the submicron white particles in the as-cast a) 0.02Nb and b) 0.04Nb steels, and the EDS analysis in c) and d) is from Spot 1 in a) and Spot 2 in b), respectively.

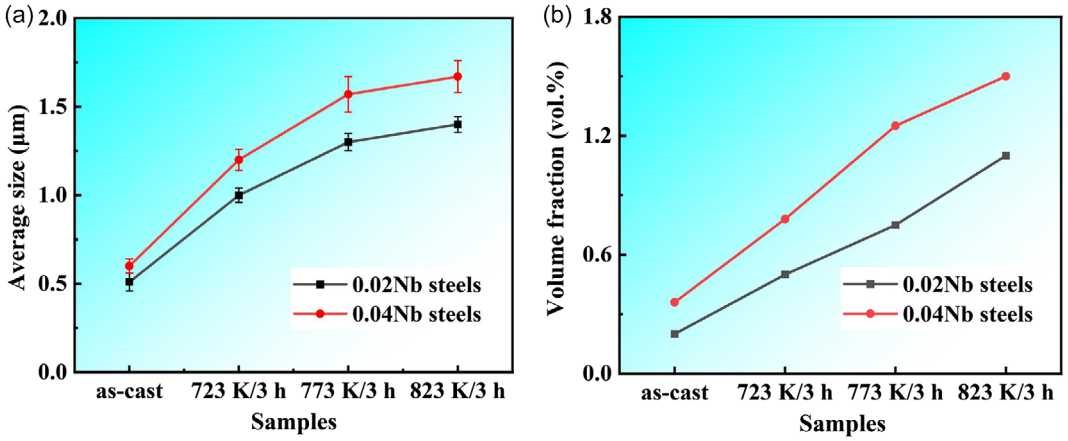


Figure 4. a) The average size and b) volume fraction of NbC particles in Nb-containing steels.

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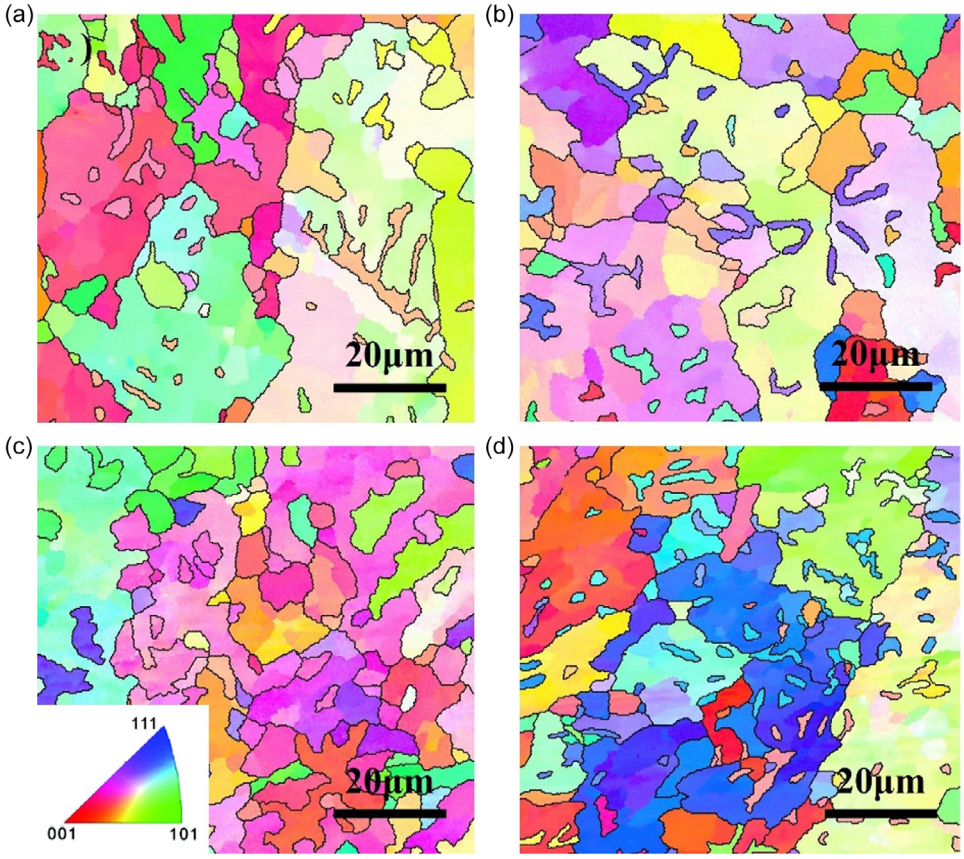


Figure 5. EBSD orientation color maps of the Nb-containing steels: a) as-cast 0.02Nb steel, b) aged 0.02Nb steel at 823 K for 3 h, c) as-cast 0.04Nb steel, and d) aged 0.04Nb steel at 823 K for 3 h.

Furthermore, the black lines in the EBSD maps are the grain boundaries identified by the minimum misorientation angle of 15°, which were used to distinguish different austenite grains. According to the analysis performed by Channel 5 software, the average size of austenite grain is 18 μm in the as-cast 0.02Nb steel, while it is determined to be 9 μm in the as-cast 0.04Nb steel. Noticeably, the austenite grain size remains almost unchanged after aging treatment at 823 K for 3 h, as depicted in Figure 3b,d. In addition, the average size of ferrite grain is measured as 3.6 μm in the as-cast 0.02Nb steel, while it is determined to be 3.1 μm in the as-cast 0.04Nb steel.

To investigate the precipitation behaviors of intragranular κ-carbide in Nb-containing steels, TEM characterization was conducted on the steels in various aging states. **Figure 6** presents the TEM dark field (DF) images of intragranular κ-carbides in austenite for the 0.02Nb and 0.04Nb steels, as well as the corresponding SAED patterns. It can be observed that a large amount of nano-scale κ-carbides are dispersed in the austenite. The coherent relationship between the ordered κ-carbides and austenite is demonstrated by the SAED analysis. In order to clearly count the size of κ-carbides, a portion of the TEM DF image was enlarged and only clear areas were selected to count the κ-carbide size by ImageJ software. For each sample, multiple regions were selected for statistical analysis of κ-carbide size, and finally took the average value. The measured content and size of κ-carbides are shown in **Figure 7**. The aging treatment increases the content and size of κ-carbides, with a higher aging temperature resulting in a greater amount of κ-carbides. After the aging treatment at 823 K, the volume fraction of κ-carbides in 0.02Nb and 0.04Nb steels increases from 3.3 and 1.6 vol% in the as-cast state to 27.8 and 26.7 vol%, respectively. Furthermore, the average κ-carbide size in 0.02Nb and 0.04Nb steels also increases from 2.0 and 1.9 nm to 7.8 and 6.4 nm, respectively. Notably, the κ-carbides in 0.04Nb steels consistently have a smaller content and size compared to those in the 0.02Nb steels.

3.1.2. Mechanical Properties

Figure 8 shows the microhardness evolution of 0.02Nb and 0.04Nb steels. As the aging temperature increases, the microhardness of both 0.02Nb and 0.04Nb steels gradually increases. For the as-cast and aged 0.02Nb steels at 723, 773, and 823 K for 3 h, the average microhardness values are measured as 406, 440, 455, and 467 HV, respectively, while that of the 0.04Nb steels are measured as 408, 452, 465, and 477 HV, respectively. The microhardness of the 0.04Nb steels consistently surpasses that of the 0.02Nb steels in the same state.

The engineering stress–strain curves of the Nb-containing steels in various aging states and the specific mechanical properties are shown in **Figure 9** and **Table 2**. As the aging

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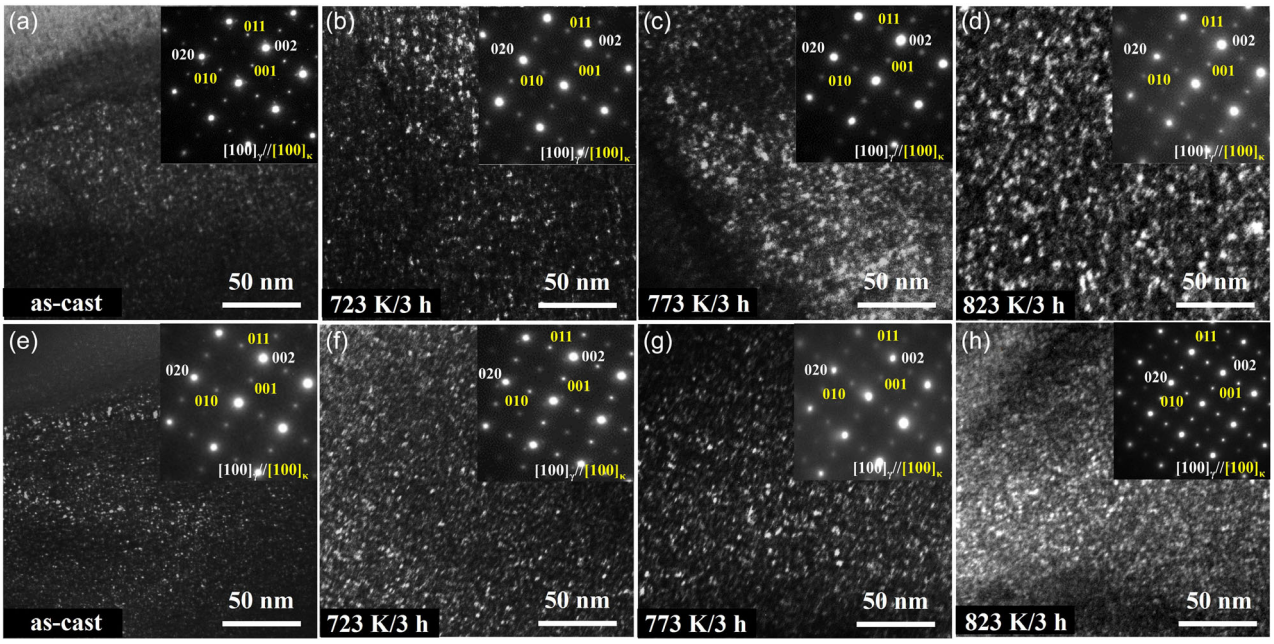


Figure 6. TEM characterization on intragranular κ-carbides in the a–d) 0.02Nb and e–h) 0.04Nb steels.

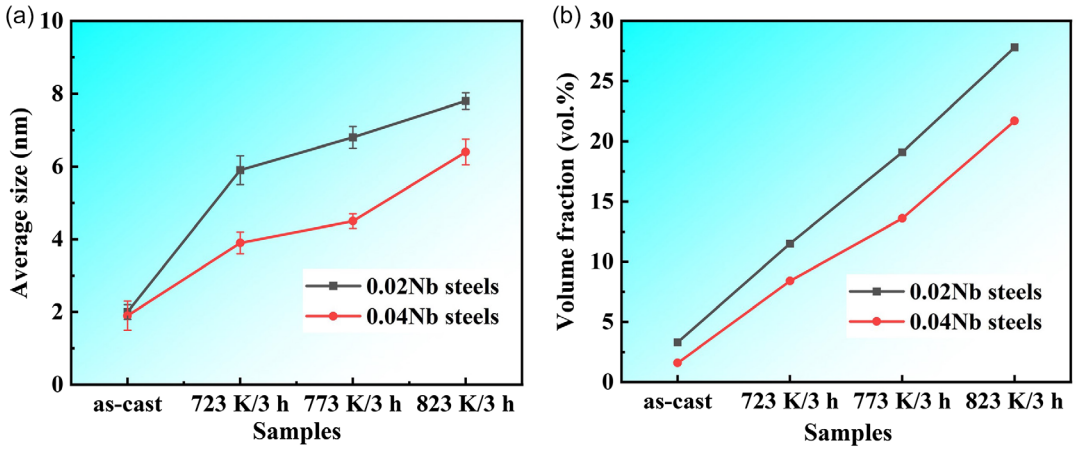


Figure 7. a) The average size and b) volume fraction of κ-carbides in the Nb-containing steels.

temperature increases, the yield strength and tensile strength of both the 0.02Nb and 0.04Nb steels gradually increase, while their ductility decreases. Following the aging process at 823 K, the 0.02Nb steel exhibits an increase in the yield strength from 880 MPa in the as-cast state to 1041 MPa, accompanied by a decrease in total elongation from 39% to 8%. For the 0.04Nb steel aged at 823 K, it experiences a significant increase in yield strength, rising from 896 MPa in the as-cast state to 1056 MPa. Similarly, this increase in strength is also accompanied by a decrease in total elongation, which decreases from 37% to 7%. Overall, the trend of changes in mechanical properties of 0.02Nb steels and 0.04Nb steels is similar. But, the 0.04Nb steels generally exhibit a slightly higher strength and a poorer plasticity than the 0.02Nb steels.

3.2. Structure and Properties of Aged Steels for Different Periods

3.2.1. Microstructure Evolution

Based on the aforementioned experimental findings, the aging treatment has a significant influence on the microstructure evolution and mechanical properties of the Nb-containing steels. And, after aging treatment at 723 K, the low-density steels could achieve better comprehensive mechanical properties. In order to comprehensively study the effects of Nb content on the microstructure and mechanical properties of medium-Mn low-density steels, the aging treatments at 723 K for longer periods were further conducted on both the 0.02Nb and 0.04Nb steels.

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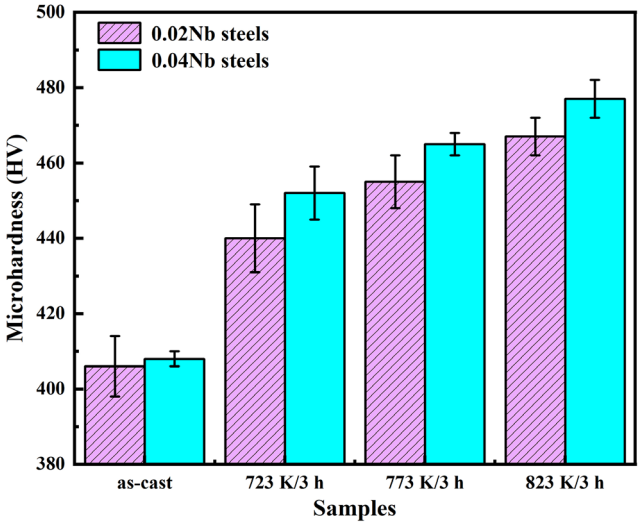


Figure 8. Microhardness evolution of the Nb-containing steels.

Figure 10 shows XRD patterns of the 0.02Nb and 0.04Nb steels aged at 723 K for 3, 6, and 9 h. Despite the alteration in aging period, the primary phases in the Nb-containing steels remain ferrite and austenite, without observable new diffraction peak in the XRD patterns.

Figure 11 exhibits the SEM images of aged 0.02Nb and 0.04Nb steels at 723 K for 6 and 9 h. As the aging time prolongs, the phase boundaries between ferrite and austenite in the Nb-containing steels become irregular, featuring numerous irregular bulges. But, there is no significant change in the volume fraction of constituent phases in the 0.02Nb and 0.04Nb steels. The ferrite contents in the 0.02Nb steels aged at 723 K for 6 and 9 h are measured as 17.1 and 18.9 vol%, respectively. Similarly, the ferrite contents in the 0.04Nb steels are 17.5 and 18.8 vol% for the same aging period. Additionally, compared to 0.02Nb steels, the size of NbC particles in 0.04Nb steels significantly increases with the prolongation of aging period.

Figure 12 shows the TEM DF images of κ-carbides in the aged 0.04Nb steels at 723 K for 6 and 9 h. Following the aging treatments for 6 and 9 h, the volume fraction of κ-carbides increases to 18.7 and 19.5 vol%, respectively. Meanwhile, their average size increases to 7 and 9 nm, respectively. Evidently, prolonging aging treatment time promotes the κ-carbide growth in the Nb-containing steels.

3.2.2. Mechanical Properties

The engineering stress–strain curves of the aged Nb-containing steels at 723 K for different periods are shown in **Figure 13**, and the specific mechanical properties are summarized in **Table 3**.

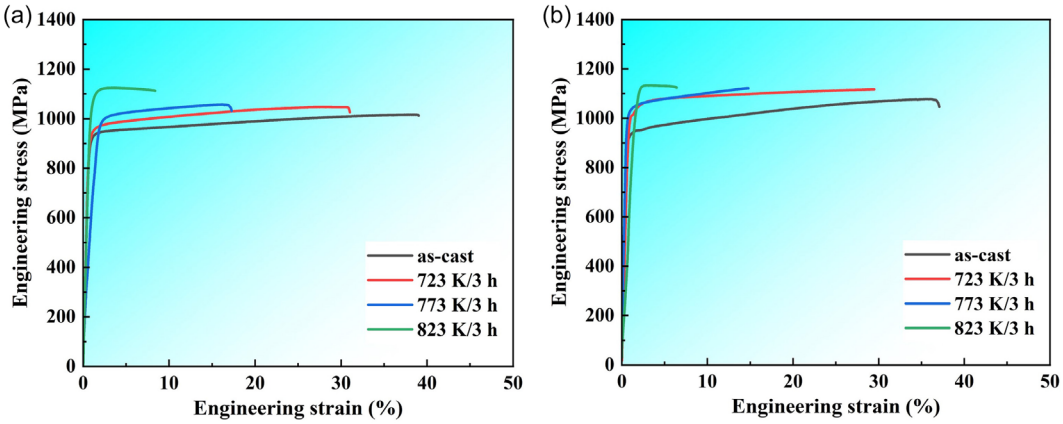


Figure 9. Engineering stress–strain curves of the a) 0.02Nb and b) 0.04Nb steels.

Table 2. Tensile mechanical properties of the Nb-containing steels.

Samples	Aging states	Yield strength [MPa]	Tensile strength [MPa]	Total elongation [%]
0.02Nb steels	as-cast	880 ± 19	1016 ± 12	39 ± 2
	723 K/3 h	912 ± 11	1047 ± 13	31 ± 3
	773 K/3 h	975 ± 18	1057 ± 21	17 ± 3
	823 K/3 h	1041 ± 15	1125 ± 10	8 ± 3
0.04Nb steels	as-cast	896 ± 23	1078 ± 18	37 ± 2
	723 K/3 h	983 ± 13	1117 ± 12	28 ± 2
	773 K/3 h	1014 ± 16	1122 ± 14	14 ± 2
	823 K/3 h	1056 ± 28	1133 ± 33	7 ± 1

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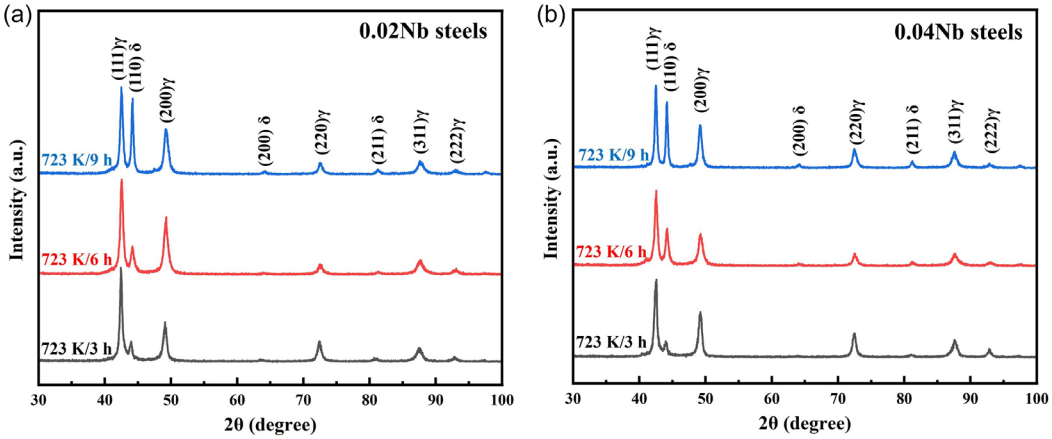


Figure 10. XRD patterns of the a) 0.02Nb steels and b) 0.04Nb steels, aged at 723 K for different periods.

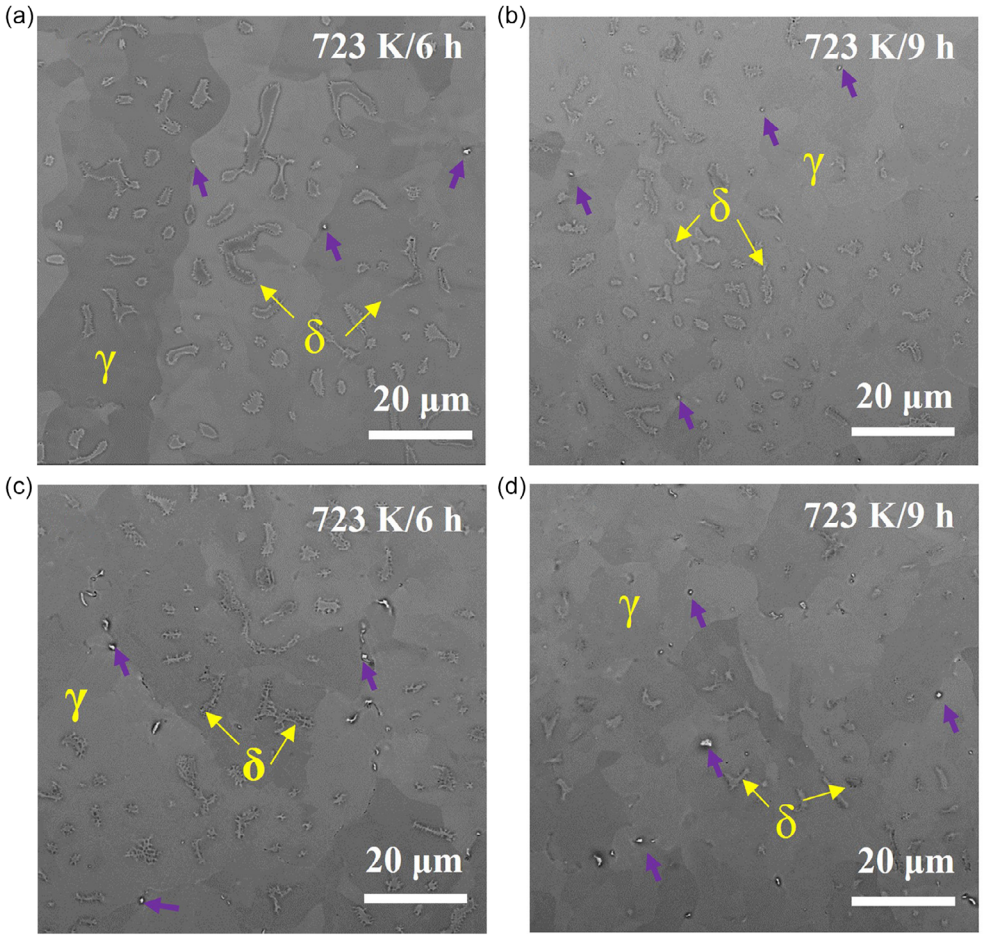


Figure 11. a,b) SEM images of 0.02Nb and c,d) 0.04Nb steels aged at 723 K for different periods.

For both the 0.02Nb and 0.04Nb steels, the aging treatment at 723 K for 6 h could result in good comprehensive mechanical properties. The yield strength of the 0.02Nb and 0.04Nb steels is increased to 950 and 1010 MPa, respectively. When the aging period was extended to 9 h, the strength of steels no longer increases, and the plasticity decreases significantly. The total elongation of the aged 0.02Nb and 0.04Nb steels for 9 h decreases to 8% and 5%, respectively. With the extension of aging time, the strength of 0.04Nb steels is still higher than that of 0.02Nb steels, but its total elongation is lower.

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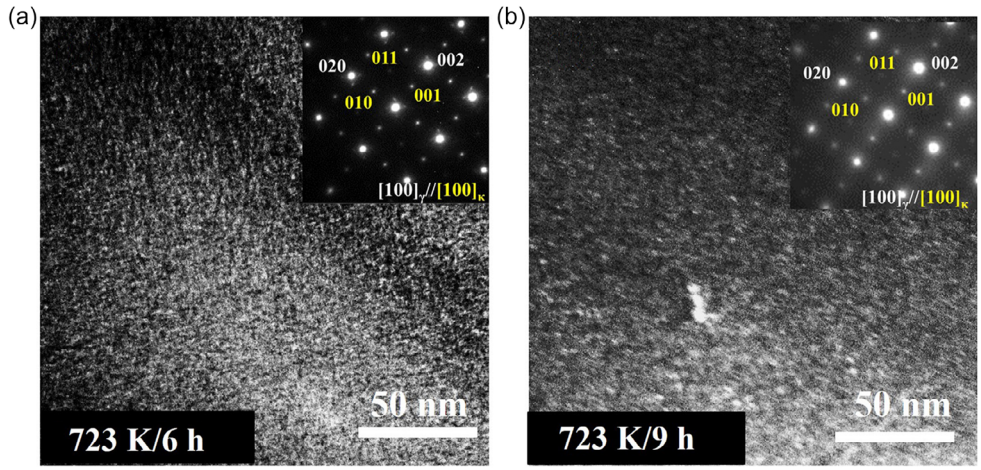


Figure 12. TEM characterization on intragranular κ-carbides in the aged 0.04Nb steels at 723 K for a) 6 h and b) 9 h.

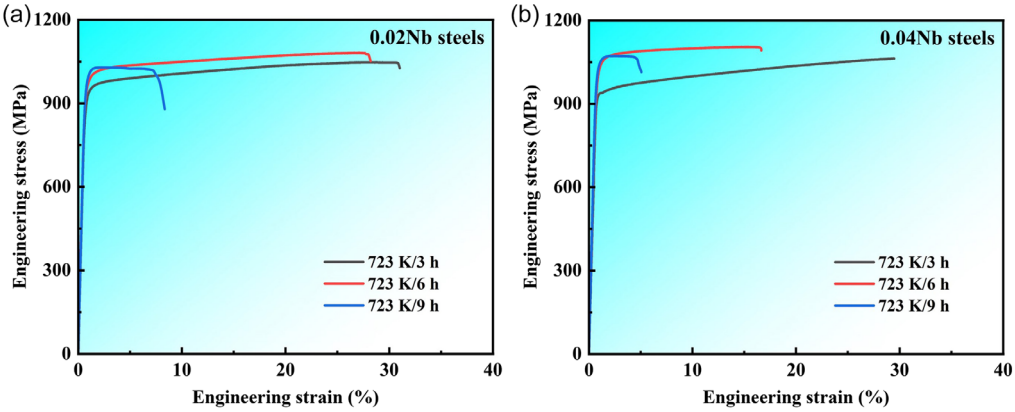


Figure 13. Engineering stress–strain curves of the a) 0.02Nb and b) 0.04Nb steels aged at 723 K for different periods.

Table 3. Tensile mechanical properties of Nb-containing steels aged at 723 K for different periods.

Samples	Aging states	Yield strength [MPa]	Tensile strength [MPa]	Total elongation [%]
0.02Nb steels	723 K/6 h	950 ± 14	1081 ± 18	28 ± 1
	723 K/9 h	975 ± 5	1029 ± 4	8 ± 2
0.04Nb steels	723 K/6 h	1010 ± 4	1104 ± 5	17 ± 1
	723 K/9 h	1019 ± 3	1072 ± 3	5 ± 1

4. Discussion

4.1. Microstructure Evolution of Nb-Containing Steel

Through near-rapid solidification and subsequent cooling, the retention of metastable austenite at room temperature has been observed in Figure 1 and 2. To examine the impact of Nb addition on the microstructural evolution, the equilibrium phase diagram of Fe-12Mn-9Al-1.4C-3Cr-xNb (wt%, $x = 0$–0.05) system was analyzed using Thermo-Calc software with TCFE10 database, as displayed in **Figure 14**. Previous studies have shown a strong correlation between the constituent phase in near-rapidly solidified Fe-Mn-Al-C low-density steel and those at the end of equilibrium solidification.[30] According to Figure 14, the Nb content has limited impact on the temperature at which the liquid-to-solid phase transformation occurs, leading to similar constituent phase content in the 0.02Nb and 0.04Nb steels.

Noteworthy, the increase in Nb content is found to refine the austenite grain size, as shown in Figure 5. This could be attributed to more intergranular NbC particles in the 0.04Nb steels, which results in a greater hindrance effect on the austenite grain boundary motion and grain growth. Based on the equilibrium phase diagram shown in Figure 14, the small changes in Nb content have a significant impact on the formation of NbC (MC phase in the diagram). As the Nb content increases from 0 to 0.05 wt%, the precipitation temperature of NbC increases from ≈973 to 1573 K. Due to the high formation temperature of NbC, the NbC particles could pin the movement of the austenite grain boundaries during the transformation of δ-ferrite → γ-austenite, hindering the growth of austenite grains.

More importantly, the increased Nb content in the low-density steels leads to a decrease in both the content and size of κ-carbides, as shown in Figure 6 and 7. To investigate the impact

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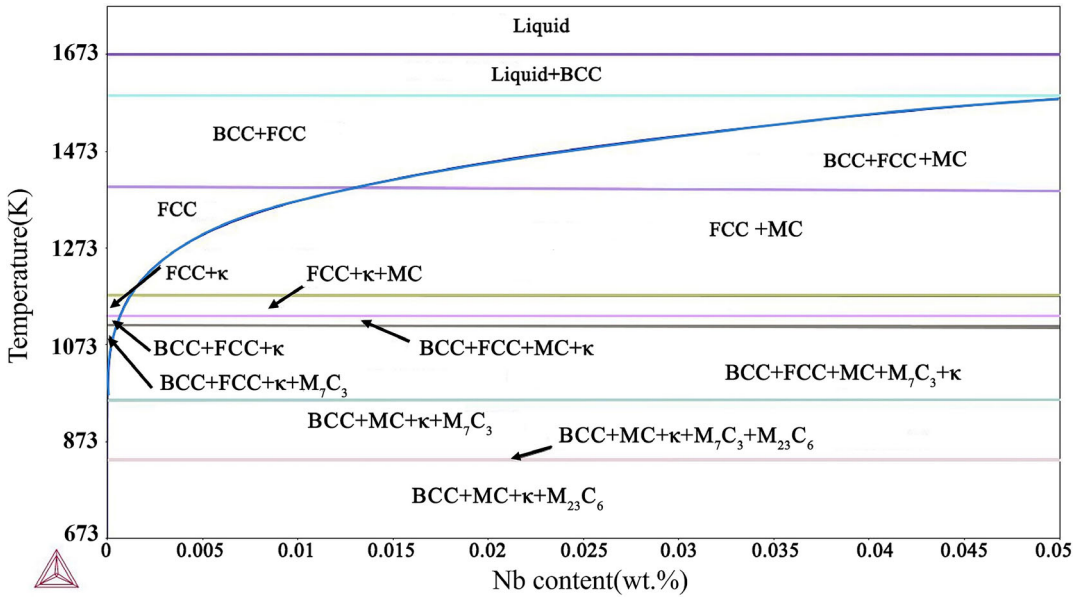


Figure 14. Equilibrium phase diagram calculated by Thermo-Calc software for Fe-12Mn-9Al-1.4C-3Cr-xNb ($x = 0$–0.05) system.

of Nb element on the κ-carbide precipitation, it is necessary to first take into account the chemical composition evolution of austenite. The chemical compositions of austenite matrix in the as-cast Nb-containing steels were analyzed using EPMA, as shown in **Figure 15** and **Table 4**. It can be found that the Nb element is uniformly dispersed within the austenite grain, with only sporadic Nb enrichment at grain boundaries in the Nb-containing steels. This indicates that the slight variations in ferrite and NbC content have minimal impact on the chemical compositions of austenite in the 0.02Nb and 0.04Nb steels, particularly for the crucial constituents of κ-carbide, namely Al and C.

Considering the uniform distribution of κ-carbides and Nb element in the austenite (Figure 6 and 15), it is reasonable to infer that Nb atoms are present within the κ-carbides. To investigate the impact of Nb atoms on the formation of κ-carbides, a

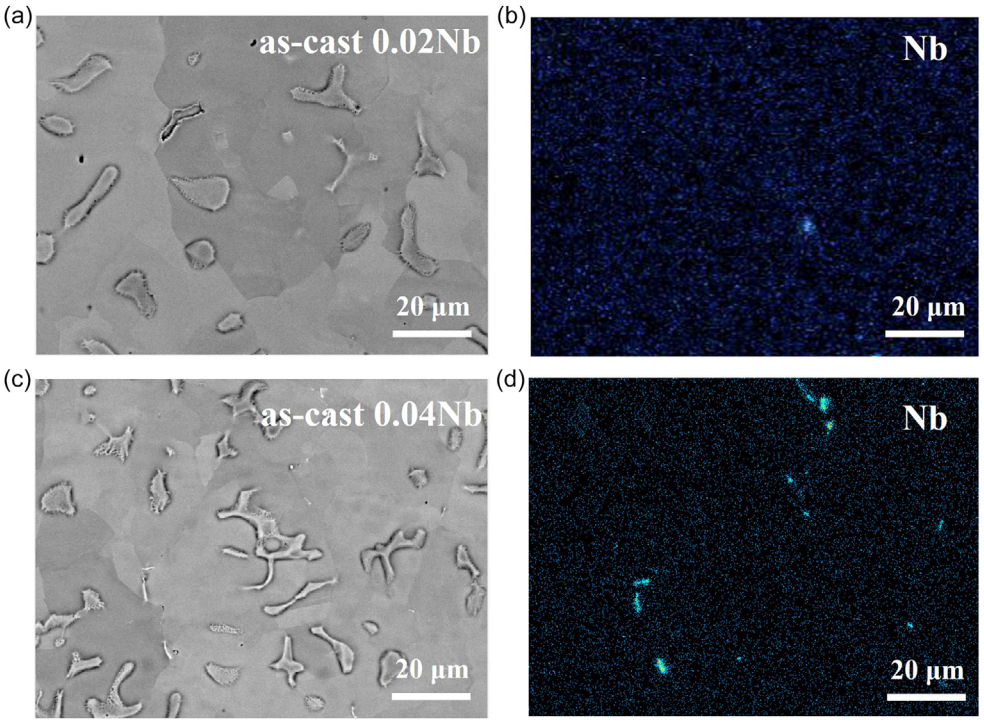


Figure 15. Microstructure morphology and chemical composition analysis by EPMA on a,b) the as-cast 0.02Nb and c,d) 0.04Nb steels.

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Table 4. The quantitative chemical compositions (wt%) of austenite in the 0.02Nb and 0.04Nb steels.

Samples	Mn	Al	Cr	C	Nb	Fe
0.02Nb steel	13.12	8.35	4.06	1.39	0.019	63.79
0.04Nb steel	13.03	8.37	3.97	1.33	0.038	64.07

$2 \times 2 \times 2$ atomic supercell structure was utilized to model the κ-carbide (Fe_3AlC, E21 structure) for first-principles calculations, as depicted in **Figure 16**a. For the inclusion of Nb atoms, two different positions were considered: the face-centered position of the cube (corresponding to the Fe position in Fe3AlC), as depicted in Figure 16b, and the corner position of the cube (corresponding to the Al position in Fe3AlC), as shown in Figure 16c. The formation energy of the three κ-carbide structures ($Fe_{24}Al_8C_8$, $Fe_{23}NbAl_8C_8$, and $Fe_{24}Al_7NbC_8$) was calculated using equation:

$$E_F = E_{DFT} - \sum N_X \mu_X \quad (1)$$

where E_F represents the formation energy, calculated as the total energy (E_{DFT}) derived from DFT, minus the sum of the product of the number of atoms of each element X (N_X) and their respective atomic chemical potentials (μ_X). The lower E_F value indicates greater stability. In this study, the formation energy for the structures of $Fe_{24}Al_8C_8$, $Fe_{23}NbAl_8C_8$, and $Fe_{24}Al_7NbC_8$ was calculated as 0.891, 2.826, and 1.453 eV, respectively. Remarkably, doping κ-carbide with Nb atoms significantly increases its formation energy.

Furthermore, with increasing the aging temperature, the average size of κ-carbides in the 0.02Nb steels is consistently slightly larger than that in the 0.04Nb steels. The diffusion coefficients of Al and C in austenite for the 0.02Nb and 0.04Nb steels were calculated using Thermo-Calc software in conjunction with the TCFE10_MOBFE5 database within the temperature range of 673–873 K. These diffusion coefficients are depicted in **Figure 17**a,b. The presence of Nb atoms slightly decreases the diffusion coefficient of C in austenite due to the strong affinity between Nb and C. On the contrary, the diffusion coefficient of Al experienced a slight increase. It has been reported that for the growth and coarsening of κ-carbides, the long-range diffusion of Al atoms had a minimal impact, while the diffusion of C atoms was more significant.[31] Therefore, the slower C diffusion caused by increased Nb content would decelerate the nucleation and growth of κ-carbides.

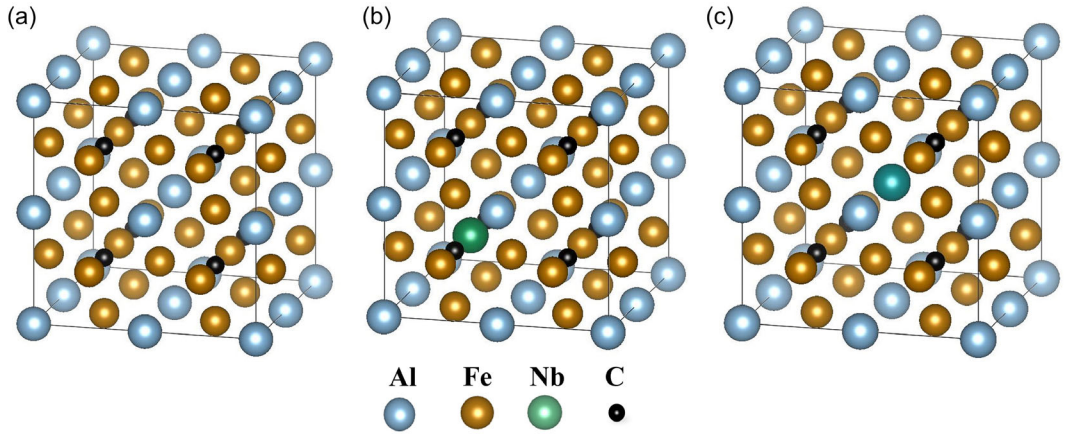


Figure 16. Calculated supercell structures of κ-carbides: a) $Fe_{24}Al_8C_8$; b) $Fe_{23}NbAl_8C_8$; c) $Fe_{24}Al_7NbC_8$.

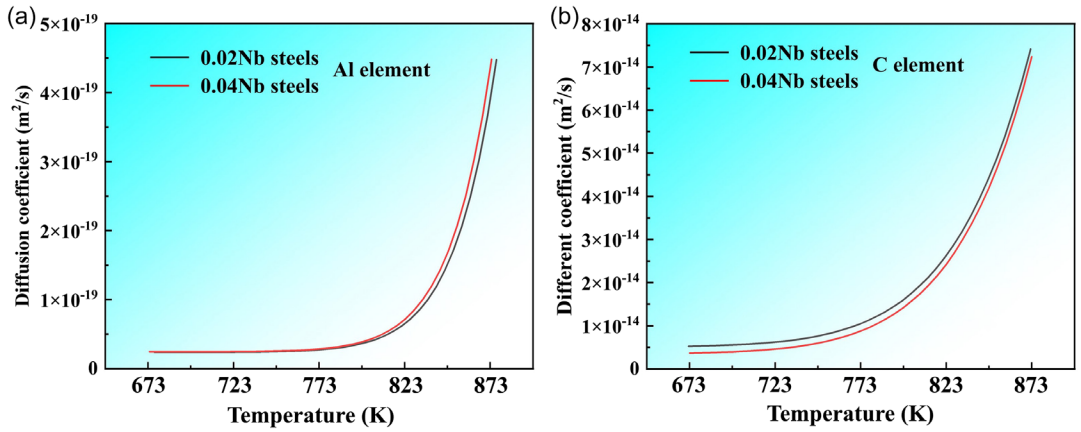


Figure 17. Calculated diffusion coefficients of a) Al and b) C in austenite of the 0.02Nb and 0.04Nb steels.

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4.2. Strengthening Mechanisms of the Nb-Containing Low-Density Steels

The uniaxial tensile tests reveal that enhancing Nb content and employing aging treatment could increase the yield strength, as evidenced by Figure 9 and Table 2. As the austenite accounts for over 80 vol% volume fraction, it plays a dominant role in determining the mechanical properties of low-density steels. Consequently, a quantitative analysis is carried out to assess the strengthening mechanisms in austenite for the 0.02Nb and 0.04Nb steels. The contribution of austenite to the yield strength of steels can be assessed as follows:

$$\sigma_a = (\sigma_0 + \sigma_s + \sigma_g + \sigma_p + \sigma_d) * V_a \tag{2}$$

where σ_0 represents lattice friction stress, σ_s, σ_g, σ_p, and σ_d represent the contributions of solid solution strengthening, grain boundary strengthening, precipitation strengthening and dislocation strengthening, respectively.

In the present work, the increased Nb content and the aging treatments exhibit minimal impact on the constituent phase fraction and its chemical compositions, as depicted in Figure 1 and 2, and Table 4. Although increasing the Nb content facilitates the formation of NbC particles in austenite, the effect on solid solution strengthening is limited, attributed to the relatively small addition content of Nb. As a result, the increased Nb content and aging treatment have little impact on the solid solution strengthening in austenite.

According to our previous work, the formation of submicron-sized particles and aging treatment would influence the dislocation density within austenite significantly.[32] The dislocation strengthening in austenite of the Nb-containing steels can be estimated by the following equation:[33]

$$\sigma_d = M\alpha G b\sqrt{\rho} \tag{3}$$

where the Taylor factor M, which reflects the grain orientation in a random distribution, is set as 3.06.[34] The material constant α is given as 0.23,[35] the Burgers vector b is 0.26,[34] the shear modulus G for austenite is 70 GPa.[34] And ρ stands for the dislocation density, which is determined using the Williamson–Hall equation:[36,37]

$$\rho = \frac{2\sqrt{3}\varepsilon}{db} \tag{4}$$

which incorporates the microscopic strain ε and the average grain size d. To acquire the value of ε, the initial step involves obtaining the half-peak width β and the related Bragg angle θ for the XRD diffraction peaks. The value of β can be obtained using the equation:[38,39]

$$\beta = 4\varepsilon \tan\theta + \frac{K\lambda}{d\cos\theta} \tag{5}$$

Among them, K represents a constant and λ signifies the wavelength of Cu Kα1 ray. Equation (5) could be simplified as:

$$\beta\cos\theta = \frac{K\lambda}{d} + 4\varepsilon\sin\theta \tag{6}$$

The value of ε in the formula is calculated from the slope of the linear fitting graph of $\beta\cos\theta$–$4\sin\theta$. The corresponding values of θ, β, $\beta\cos\theta$, and $4\sin\theta$ for austenite in the 0.02Nb are recorded in **Table 5**, and the values of ε and ρ for austenite are shown in **Table 6**. Based on the aforementioned data, the dislocation strengthening in austenite for the as-cast 0.02Nb steel is calculated to be 113.1 MPa. After aging at 723, 773, and 823 K, the dislocation strengthening of austenite in the 0.02Nb steels is 87.4, 70.8, and 40.2 MPa, respectively. For the austenite in 0.04Nb steels, the dislocation strengthening is 169.8 MPa in the as-cast state. After aging at 723, 773, and 823 K, the dislocation strengthening in austenite is 129.0, 106.1, and 98.9 MPa, respectively.

The increased Nb content refines the austenite grain, as evidenced in Figure 5. The Hall–Petch formula was used to calculate the grain boundary strengthening (σ_g) of austenite:[40]

$$\sigma_g = k_\gamma d^{-\frac{1}{2}} \tag{7}$$

Table 5. Values of θ, β, $\beta\cos\theta$, and $4\sin\theta$ for austenite in 0.02Nb and 0.04Nb steels.

Peaks	Aging states	0.02Nb				0.04Nb			
		Θ	β	$\beta\cos\theta$	$4\sin\theta$	Θ	β	$\beta\cos\theta$	$4\sin\theta$
(111)	as-cast	21.242	0.409	0.381	1.449	21.242	0.373	0.348	1.449
	723 K	21.267	0.419	0.390	1.451	21.217	0.422	0.393	1.448
	773 K	21.217	0.428	0.399	1.448	21.267	0.418	0.390	1.451
	823 K	21.192	0.406	0.379	1.446	21.217	0.428	0.399	1.448
(200)	as-cast	24.628	0.577	0.525	1.667	24.603	0.605	0.550	1.665
	723 K	24.628	0.572	0.520	1.667	24.553	0.578	0.526	1.662
	773 K	24.552	0.572	0.520	1.662	24.628	0.563	0.512	1.667
	823 K	24.603	0.608	0.553	1.665	24.578	0.572	0.520	1.664
(220)	as-cast	36.253	0.637	0.514	2.365	36.228	0.691	0.557	2.364
	723 K	36.303	0.63	0.508	2.368	36.253	0.642	0.518	2.365
	773 K	36.253	0.601	0.485	2.365	36.303	0.604	0.487	2.368
	823 K	36.253	0.560	0.451	2.364	36.228	0.601	0.485	2.364

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Table 6. Values of ε and ρ for austenite in 0.02Nb and 0.04Nb steels.

Samples	Aging states	ε	ρ [m^{-2}]
0.02Nb steels	as-cast	0.15	1.11×10^{14}
	723 K/3 h	0.09	6.66×10^{13}
	773 K/3 h	0.06	4.44×10^{13}
	823 K/3 h	0.02	1.48×10^{13}
0.04Nb steels	as-cast	0.17	2.51×10^{14}
	723 K/3 h	0.10	1.48×10^{14}
	773 K/3 h	0.07	1.03×10^{14}
	823 K/3 h	0.06	8.88×10^{13}

where k_y is the Hall–Petch coefficient (14.55 MPa • $mm^{-1/2}$[41]), and d represents the average grain size of the austenite. As the grain size remains unchanged with the aging treatment, the grain boundary strengthening also remains constant. The grain boundary strengthening for the as-cast 0.02Nb steel is calculated as 90.8 MPa. After aging at 723, 773, and 823 K, the grain boundary strengthening in austenite is 90.6, 89.9, and 88.6 MPa, respectively. For the austenite in 0.04Nb steels, the grain boundary strengthening is 128.4 MPa in the as-cast state. After aging at 723, 773, and 823 K, the grain boundary strengthening in austenite is 127.0, 125.8, and 125.6 MPa, respectively.

The precipitation of NbC particles and κ-carbides is greatly influenced by both the Nb content and the aging treatments, as illustrated in Figure 4 and 7. Notably, due to the small size, the nano-scale κ-carbides observed in this study could be sheared by dislocations under plastic deformation.[34,42] The strength enhancement attributed to the shearing of κ-carbides can be quantified by:[42]

$$\sigma_p = \frac{M}{N}\frac{\gamma_{APB}}{b}\sqrt{V_f}\left[\sqrt{\frac{12\gamma_{APB}r}{\pi G b^2}} - \sqrt{V_f}\right] \quad (8)$$

where N represents the number of accumulated dislocations aiding in shearing intragranular κ-carbides, γ_{APB} stands for antiphase boundary energy, V_f represents the content of κ-carbides, and r represents the κ-carbide average size. Considering the small size of intragranular κ-carbides, the values of N and γ_{APB} were set as 6 and 350 mJ m^{-2}, respectively.[42] By utilizing Equation (8) along with the data obtained from Figure 7, the precipitation strengthening attributed to the κ-carbides in the as-cast 0.02Nb steel is calculated as 59.4 MPa. After aging at 723, 773, and 823 K, the precipitation strengthening by κ-carbide are 180.6, 230.0, and 274.8 MPa, respectively. For the 0.04Nb steels under the same aging condition, the strengthening contribution from κ-carbide precipitation is estimated as 53.1, 127.4, 156.8, and 230.2 MPa, respectively.

Due to the impermeable nature of large-size NbC particles, the strength contribution of these particles can be attributing to the Orowan looping mechanism. It could be assessed using the Ashby–Orowan equation:[43]

$$\sigma_{oro} = \left(\frac{0.538Gbf^{\frac{1}{2}}}{D}\right) \times \left(\ln\frac{D}{2b}\right) \quad (9)$$

The value of f represents the content of NbC particles, while D denotes their average size. Accordingly, the calculated strength contribution by the precipitation strengthening of NbC particles for the as-cast and aged 0.02Nb steels at 723, 773, and 823 K is 4.9, 4.3, 4.2, and 4.7 MPa, respectively. For the 0.04Nb steels, the precipitation strengthening by NbC particles is 6.2, 4.7, 4.6, and 4.8 MPa, respectively

In addition, the strength contribution of ferrite for the 0.02Nb and 0.04Nb steels should also be analyzed in detail. First, considering the little effect of the change in Nb content and aging treatment on the chemical composition of austenite (Table 4) and the volume fraction of ferrite (Figure 2), it can be inferred that the chemical composition of ferrite remains almost unchanged. That is to say, the increase in Nb content has a limited effect on the solid solution strengthening in ferrite. Meanwhile, the similar ferrite grain sizes in different samples indicate that the grain boundary strengthening is almost unaffected by Nb content and aging treatment. In addition, according to the equilibrium phase diagram in Figure 14, the increase of 0.02 wt% Nb content would not lead to significant phase transformation and strengthening phase precipitation in ferrite. For the dislocation strengthening in ferrite of the Nb-containing steels, it can be estimated by Equation (3). The Taylor factor M_F, which reflects the grain orientation in a random distribution, is set as 2.[44] The material constant α_F is given as 0.4,[44] the shear modulus G_F for ferrite is 76 GPa.[44] The corresponding values of θ, β, $\beta\cos\theta$, and $4\sin\theta$ for ferrite in the 0.02Nb are recorded in **Table 7**, and the values of ε and ρ for ferrite are shown in **Table 8**. Considering the volume fraction of ferrite in steels, the dislocation strengthening in ferrite for the as-cast 0.02Nb steel is calculated to be 79.7 MPa. After aging at 723, 773, and 823 K, the dislocation strengthening of ferrite in the 0.02Nb steels is 84.0, 83.5, and 93.3 MPa, respectively. For the ferrite in 0.04Nb steels, the dislocation strengthening is 84.3 MPa in the as-cast state. After aging at 723, 773, and 823 K, the dislocation strengthening in ferrite is 92.2, 96.4, and

Table 7. Values of θ, β, $\beta\cos\theta$, and $4\sin\theta$ for ferrite in 0.02Nb and 0.04Nb steels.

Peaks	Aging states	0.02Nb				0.04Nb			
		Θ	β	$\beta\cos\theta$	$4\sin\theta$	Θ	β	$\beta\cos\theta$	$4\sin\theta$
(110)	as-cast	21.960	0.424	0.393	1.496	21.937	0.403	0.373	1.494
	723 K	21.970	0.462	0.428	1.497	22.014	0.434	0.402	1.499
	773 K	22.453	0.428	0.395	1.528	22.073	0.418	0.387	1.503
	823 K	21.983	0.457	0.424	1.497	21.980	0.430	0.398	1.497

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Table 8. Values of ε and ρ for ferrite in 0.02Nb and 0.04Nb steels.

Samples	Aging states	ε	ρ [m^{-2}]
0.02Nb steels	as-cast	0.262	9.69×10^{14}
	723 K/3 h	0.285	1.05×10^{15}
	773 K/3 h	0.258	9.55×10^{14}
	823 K/3 h	0.283	1.04×10^{15}
0.04Nb steels	as-cast	0.249	1.07×10^{15}
	723 K/3 h	0.268	1.15×10^{15}
	773 K/3 h	0.257	1.10×10^{15}
	823 K/3 h	0.265	1.13×10^{15}

96.1 MPa, respectively. It can be seen that the increase in Nb content and aging heat treatment have limited effects on the strength contribution of ferrite.

Table 9 shows the calculated strength contributions from various reinforcement factors in the austenite of low-density steels. With increasing the aging temperature, the effect of dislocation strengthening is reduced, while the effect of κ-carbide precipitation strengthening increases, exceeding the reduction in dislocation strengthening. Moreover, the aging treatment facilitates the formation of NbC particles in austenite, decreasing the total elongation of steels. It is observed that although the κ-carbide precipitation strengthening effect is more pronounced in the 0.02Nb steels, the effects of dislocation strengthening and grain boundary strengthening are more significant in the 0.04Nb steels. Consequently, the 0.04Nb steels exhibit higher yield strength.

Moreover, extended aging treatment at 723 K does not markedly alter the yield strength of Nb-containing steels, as depicted in Figure 13. This can be attributed to the fact that although the precipitation strengthening by κ-carbide intensifies, while the grain boundary strengthening remains relatively unchanged with extended aging and the dislocation strengthening diminishes. Consequently, the extension of aging time has only a minor impact on the strength of Nb-containing steels. The 0.04Nb steel aged at 723 K for 3 h exhibits the most favorable comprehensive mechanical properties.

Table 9. Theoretical calculation results of various strengthening effects in the austenite of low-density steels.

Samples	Aging states	σ_d [MPa]	σ_g [MPa]	σ_p [MPa]	σ_{oro} [MPa]
0.02Nb steels	as-cast	113.1	90.8	59.4	4.9
	723 K/3 h	87.4	90.6	180.6	4.3
	773 K/3 h	70.8	89.9	230.0	4.2
	823 K/3 h	40.2	88.6	274.8	4.7
0.04Nb steels	as-cast	169.8	128.4	53.1	6.2
	723 K/3 h	129.0	127.0	127.4	4.7
	773 K/3 h	106.1	125.8	156.8	4.6
	823 K/3 h	98.9	125.6	230.2	4.8

5. Conclusions

This study investigated the effects of Nb content on microstructure evolution and mechanical properties of the medium-Mn Fe-12Mn-9Al-3Cr-1.4C-0.02/0.04Nb (wt%) low-density steels. The following are the main findings that can be summarized: 1) Both the 0.02Nb and 0.04Nb steels consisted of ferrite and austenite, with a ferrite content of ≈16 vol%. The increase of Nb content promoted the formation of NbC particles, thereby refining the austenite grain size significantly. 2) The Nb element was found to be detrimental to the formation and growth of κ-carbides, which was attributed to the increased formation energy of κ-carbide by Nb doping. Because of this, the aging treatment at 723–823 K could significantly promote the growth of κ-carbides in Nb-containing steels, but no coarsened intergranular κ-carbides were observed. 3) Compared to the 0.02Nb steels, the addition of 0.04 wt% Nb resulted in the yield strength improvement of ≈71 MPa. Based on the strengthening mechanism analysis, the increase in Nb content enhanced the dislocation strengthening and grain refinement strengthening, while it also reduced the κ-carbide content in the austenite, thereby weakening the precipitation strengthening effect in the steels. 4) With increasing the aging temperature, the dislocation density in the Nb-containing low-density steels decreased, weakening the dislocation strengthening, but the precipitation and growth of κ-carbides enhanced the precipitation strengthening in the steel significantly. For the Nb-containing steels aged at temperatures exceeding 773 K, the precipitation of κ-carbide was the principal strengthening factor.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

low-density steel, mechanical properties, nano-precipitation behavior, Nb addition

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